

Self-test Exercise

1. A dependent variable is also known as a(n) _____.
 - a. explanatory variable
 - b. control variable
 - c. predictor variable
 - d. response variable

2. In the equation $y = \beta_0 + \beta_1 x + u$, β_0 is the _____.
 - a. dependent variable
 - b. independent variable
 - c. slope parameter
 - d. intercept parameter

3. In the equation $y = \beta_0 + \beta_1 x + u$, what is the estimated value of β_0 ?
 - a. $\bar{y} - \hat{\beta}_1 \bar{x}$
 - b. $\bar{y} + \beta_1 \bar{x}$
 - c.
$$\frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i)^2}$$
 - d. $\sum_{i=1}^n xy$

4. In the equation $c = \beta_0 + \beta_1 i + u$, c denotes consumption and i denotes income. What is the residual for the 5th observation if $c_5 = \$500$ and $\hat{c}_5 = \$475$?
 - a. \$975
 - b. \$300
 - c. \$25
 - d. \$50

5. What does the equation $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ denote if the regression equation is $y = \beta_0 + \beta_1 x_1 + u$?
 - a. The explained sum of squares
 - b. The total sum of squares
 - c. The sample regression function
 - d. The population regression function

6. If x_i and y_i are positively correlated in the sample then the estimated slope is ____.

- a. less than zero
- b. greater than zero
- c. equal to zero
- d. equal to one

7. Consider the following regression model: $y = \beta_0 + \beta_1 x_1 + u$. Which of the following is a property of Ordinary Least Square (OLS) estimates of this model and their associated statistics?

- a. The sum, and therefore the sample average of the OLS residuals, is positive.
- b. The sum of the OLS residuals is negative.
- c. The sample covariance between the regressors and the OLS residuals is positive.
- d. The point $(\bar{x}, \bar{y})$ always lies on the OLS regression line.

8. The explained sum of squares for the regression function, $y_i = \beta_0 + \beta_1 x_1 + u_1$, is defined as ____.

- a. $\sum_{i=1}^n (y_i - \bar{y})^2$
- b. $\sum_{i=1}^n (\hat{y} - \bar{y})^2$
- c. $\sum_{i=1}^n \hat{u}_i$
- d. $\sum_{i=1}^n (u_i)^2$

9. If the total sum of squares (SST) in a regression equation is 81, and the residual sum of squares (SSR) is 25, what is the explained sum of squares (SSE)?

- a. 64
- b. 56
- c. 32
- d. 18

10. If the residual sum of squares (SSR) in a regression analysis is 66 and the total sum of squares (SST) is equal to 90, what is the value of the coefficient of determination?

- a. 0.73
- b. 0.55
- c. 0.27
- d. 1.2

11. In a regression equation, changing the units of measurement of only the independent variable does not affect the _____.
a. dependent variable
b. slope
c. intercept
d. error term
12. The error term in a regression equation is said to exhibit homoskedasticity if _____.
a. it has zero conditional mean
b. it has the same variance for all values of the explanatory variable
c. it has the same value for all values of the explanatory variable
d. if the error term has a value of one given any value of the explanatory variable
13. In the regression of y on x , the error term exhibits heteroskedasticity if _____.
a. it has a constant variance
b. $\text{Var}(y|x)$ is a function of x
c. x is a function of y
d. y is a function of x
14. The sample covariance between the explanatory variables and the Ordinary Least Square (OLS) residuals is always positive.
a. True
b. False
15. R^2 is the ratio of the explained variation compared to the total variation.
a. True
b. False
16. The variance of the slope estimator increases as the error variance decreases.
a. True
b. False